Airport Runway Foreign Object Debris Detection System Based on Arc-Scanning SAR Technology

Yuming Wang, *Member, IEEE*, Qian Song, Jian Wang, and Huimin Yu

***Abstract*— Due to the small size of foreign object debris (FOD) and varied complex weather conditions, the detection of FOD on airport runways is a great challenge. Radar is an important method for detecting FOD targets. However, almost all the existing systems are based on real apertures, which have disadvantages such as low azimuth resolution and susceptibility to rain interference. Here, an innovative FOD detection radar system based on arc-scanning synthetic aperture radar (AS-SAR) technology, the AS-SAR based FOD detection system (AS-FODR), achieves omnidirectional coverage with a very high azimuth resolution and the suppression of flicker clutter, such as rain drops in severe weather. According to the radar imaging simulation of a scene under rainy conditions and the information processing analysis of field experiments, a prototype system was built, and an efficient data process flow was proposed. In short, a one centimeter FOD target was detected on the runway more than 250 m away, proving that the use of AS-FODR is feasible and effective.**

***Index Terms*— Airport runway, arc-scanning synthetic aperture radar (AS-SAR), detection system, foreign object debris (FOD).**

I. Introduction

FOREIGN object debris (FOD) comprises foreign objects that may damage aircraft and equipment or threaten the safety of airport staff and passengers on the ground in active areas such as airport runways. With the rapid development of the world's civil aviation industry in recent years, FODs have been observed to puncture tires and damage aircraft from time to time, causing not only greater economic losses to airlines but also greater risks to flight safety and personnel safety [1]. At present, the majority of the world's airport FOD detection is mainly done manually, and the overall level of automation and mechanization is still low. For this reason, research on detection equipment for FODs is imminent and has received increasing amounts of attention [2], [3].

At present, FOD detection equipment mainly includes photoelectric detection equipment, radar detection equipment, and hybrid detection equipment [4]–[7]. Photoelectric detection equipment installation and maintenance costs are low, and with the development of image processing technology, the detection performance is improving and the recognition rate is gradually increasing. However, such improvements are limited by low-light conditions such as those during the night and are also vulnerable to rain, snow, and other extreme weather conditions. Radar detection equipment has a high detection rate, high positioning accuracy, and high reliability, can function all day, and has superior environmental adaptability, which is the focus of this article. However, there are some problems such as difficult identification and high false alarm rates. Hybrid detection equipment combines radar detection and photoelectric detection, demonstrating the advantages of both the detection methods.

From 2007 to 2010, the Federal Aviation Administration (FAA) conducted tests and evaluations on four kinds of detection equipment [8], including QinetiQ's Tarsier, Trex's FOD Finder, Xsight Systems' FODetect, and Stratech Systems' iFerret. Based on this study, the FAA issued the Advisory Circular on FOD detection (ac, 150/5220–24). Among these types of equipment, iFerret is a video detection system that finds FOD through high-resolution cameras [9] while Tarsier, FOD Finder, and FODetect are hybrid detection equipment that integrate radar and cameras [10]–[15].

Radar plays an important role in FOD detection and is therefore the focus of this article. Tarsier radar is installed on both sides of the airport runway and contains a frequency-modulated continuous wave (FMCW) system with a working frequency of 94 GHz with a narrow antenna beam and high resolution; its detection distance can reach 2000 m when the target size is large. FOD finder radar equipment is installed on the top of a car, with a working frequency of 78–81 GHz and detection distance of 200 m. FODetect radar is installed at a low altitude on both sides of the runway; it adopts an FMCW system and works at 77 GHz. This equipment also has the characteristics of a narrow antenna beam and high resolution, and the detection range is 60 m. In addition, Feil *et al.* [16] constructed an FMCW FOD detection system with a working frequency of 73–80 GHz, which has a high sensitivity and synchronous target detection ability and can detect small targets such as nuts. The detection range of this system is better than 110 m. The Chengdu Saying Company, Chengdu, China, developed a millimeter-wave radar FOD detection system with a separation transceiver and transmitter using an FMCW system with a bandwidth of 800 MHz in 2010. Chengdu University of Electronic Science and Technology, Chengdu, China [17],

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Yuming Wang and Huimin Yu are with the College of Information Science and Engineering, Hunan Normal University, Changsha 410081, China (e-mail: wangyuming@hunnu.edu.cn; yhm@hunnu.edu.cn).

Qian Song is with Hunan GHz Info. Tech. Company Ltd., Changsha 410005, China.

Jian Wang is with the School of Electronic Science and Engineering, National University of Defense Technology, Changsha 410073, China.

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Shanghai Jiao Tong University Shanghai, China [18], and Beijing Jiaotong University, Beijing, China [19] have also carried out exploratory research on radar-based FOD detection.

Currently, FOD radar detection equipment adopts the real aperture scanning mode and uses a 1-D range profile to detect the target through a narrow beam antenna. There are three main corresponding problems First, if high-resolution imaging is to be carried out, the antenna beam should be as narrow as possible, so it is necessary to design and process a large-aperture millimeter-wave antenna; such an antenna is difficult to process and causes a large volume and weight of the system Second, the motion error of the turntable has a large impact on imaging, and the unstable rotation accuracy reduces the direction-finding accuracy. If the turntable is equipped with high stability, the volume, weight, and cost of the system will be further improved. Third, due to the principle of real aperture scanning, the ability to suppress flicker clutter such as raindrops and splashes on the ground is poor, resulting in a serious decline in the use efficiency in bad weather. To solve the above problems, this article designs an airport FOD detection system based on arc-scanning synthetic aperture radar (AS-SAR).

Synthetic aperture radar (SAR) is a high-resolution 2-D imaging radar system that is often used in ground-based, airborne, and onboard platforms. It uses the movement of the platform to synthesize a virtual large-aperture array antenna to achieve high-resolution imaging in the azimuth. After years of research and application, SAR systems and processing methods have become very mature [20]–[22]. Considering that SAR has the capability of high-resolution earth observation and can achieve high-precision detection of tiny targets on the ground, it has great potential for FOD target detection. AS-SAR is a special SAR with a circular arc aperture that has a 360° omnidirectional observation capability. The maximum radiation direction of its antenna is perpendicular to the direction of movement. AS-SAR imaging form was first proposed by Klausing *et al.* [23], [24] of the German Aerospace Research Establishment in 1989. The imaging of a ground scene was achieved through the rotation of a helicopter rotor; since 2013, researchers from the Institute of Electronics of the Chinese Academy of Sciences, Beijing, China, the Gangwon National University, Chuncheon-si, South Korea, and the University of Florence, Florence, Italy, have applied this method to 3-D topographic mapping and deformation measurements. Additionally, scientific researchers have also carried out corresponding exploration and research in AS-SARbased 3-D imaging, elevation target imaging, and surface imaging [25]–[30].

Applying AS-SAR to airport runway FOD detection has natural advantages, which are presented in the following sentences. The first advantage is that the wide-beam antenna is small in size, so the corresponding system is small in size and weight and has good adaptability. This system can be installed on fixed towers or directly installed on the existing FOD disposal vehicle at the airport; therefore, no special vehicle is required, and the overall use cost is low The second advantage is that the antenna aperture is not related to the antenna size but rather is related to the antenna beamwidth, which can be used to achieve extremely high imaging resolution through

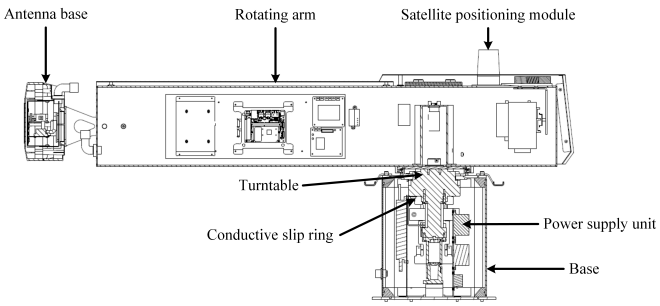


Fig. 1. Profile of the radar engineering structure.

algorithms; this is helpful for the detection of small targets The third advantage is that an AS-SAR 2-D image is generated by coherent accumulation of multiple 1-D range profiles. Compared with traditional real aperture radar, this system requires lower platform rotation accuracy and can maintain high target-direction-finding and ranging accuracies Fourth, based on the characteristics of virtual aperture imaging, the system also has a good suppression effect on the flicker clutter formed on the runway in severe weather such as rain.

The remainder of this article is organized as follows. In Section II, this article introduces the radar system design. According to the working principle of AS-SAR, a radar engineering structure and radar sensor design are presented. Section III discusses the radar signal processing flow and analyzes the signal model, imaging algorithm, and detection algorithm. Section IV simulates the system's suppression of flicker clutter under rainy conditions. Section V provides the actual system's imaging and detection results for small FOD targets on sunny and rainy days. Finally, the conclusion is given in Section VI.

II. Design of the Radar System

The radar used in this article to detect FOD adopts the AS-SAR system; its core design is that the antenna phase center uses the rotating arm to form an arc aperture and then uses the synthetic aperture principle to generate high-resolution images. Therefore, the radar system can be divided into an electromechanical subsystem and a sensor subsystem. The electromechanical subsystem is the carrier of the sensor subsystem, which provides support for the radar antenna to obtain an arc trajectory and time-sequence pulse The sensor subsystem transmits and receives the regular microwave signal and uses the corresponding algorithm to realize imaging and detection of the FOD target.

A. Radar Engineering Structure

Fig. 1 shows the profile of the radar structure. The radar structure is composed of seven parts: the base, power supply unit, turntable, rotating arm, conductive slip ring, antenna base, and satellite positioning module.

The base is used to install, store, and protect other components, also serving to fix the radar and make it work smoothly; the power supply unit is composed of alternating current–direct

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current (ac–dc) converters, batteries, etc., to provide electricity for the operation of the whole system. The turntable uses a combination of a high-speed servo motor and a deceleration mechanism to achieve low-speed and large-torque actuator input through mechanical transmission; this avoids the interval of low-speed motor vibration and accurately controls the movement process to ensure the accuracy of the rotation position and smooth trigger pulse frequency. The conductive slip ring is a rotating electrical interface that uses the sliding contact and electromagnetic coupling of the conductive parts to solve the problem related to transmitting power and data signals from the fixed structure to the rotating structure during unrestricted continuous rotation. The rotating arm is a square aluminum alloy tube installed on the turntable; it is hollow and used to install radar sensor subsystem components. The antenna base is installed at the front of the rotating arm and consists of an aluminum alloy substrate, a pitch bracket, and a glass-fiber-reinforced plastic radome; the pitch attitude of the antenna can be adjusted Finally, the satellite positioning module mainly provides timing services for the system and can also provide approximate position in the experimental test with low positioning accuracy requirement.

B. Radar Sensor

1) Frequency Selection: To meet the Federal Communications Commission (FCC) requirements and make better use of the atmospheric window with less attenuation near 94 GHz while also better detecting small FOD targets at approximately 1 cm, this system works in the W-band.

In the process of FOD detection, it is necessary to detect targets with small sizes and to separate multiple small targets at very close distances. Therefore, to ensure that AS-SAR has a high image resolution, we should also ensure that the system itself has a high range resolution. Here, we establish the design that the system bandwidth B is 1 GHz. Then, it can be calculated that the range resolution of this system is

$$\Delta r = \frac{c}{2B} = \frac{3e8}{2 \times 1e9} = 0.15 \text{ m} \tag{1}$$

where c is the speed of light.

2) Radar Sensor Architecture: Radar sensors consist of receiving and transmitting antennas, signal generation (SG) modules, radio frequency (RF) modules, timing control (TC) modules and real-time acquisition and processing (RTAAP) modules. The architecture diagram of the radar sensor, as shown in Fig. 2, displays the signal transfer process. An FMCW signal generated by the SG module is upconverted and amplified by the RF module and then fed to the transmitting antenna for radiation. After being reflected by the FOD target, the signal is transmitted to the receiving antenna and then amplified, mixed, and low-passed by the RF module. After that, the signal is collected in real-time by the RTAAP module. Finally an image is formed using the signal, which is used for detection.

The SG module is responsible for the generation of intermediate frequency (IF) signals. SG uses direct digital synthesis (DDS) to drive the phase locked loop (PLL), which not only retains the advantages of DDS low conversion time and high resolution but also achieves good noise and false performance of PLL. Moreover, the circuit structure of this scheme is simple, the output frequency and bandwidth are high, and it can achieve good phase noise and false indicators in line with the design requirements. Modern radar signals include continuous waves and pulses. This article adopts the linear FMCW signal system to avoid the near-end blind area caused by the time-sharing operation of the pulse signal system and eliminates the minimum detection distance to facilitate the flexible deployment of the actual system according to the scene. For this reason, the loss of dynamic range caused by antenna leakage should be considered in the design process for other modules.

The RF module includes a transmitter and a receiver. The transmitter needs to upconvert the IF signal, amplify it, and feed it into the transmitting antenna; the design of the receiver adopts a zero-IF structure. First, this component receives the scene echo, performs preamplification, mixes the echo with the transmitted signal for defrequency modulation and coherent reception to generate a zero-IF signal, and then processes it through a low-pass filter. Compared with the receivers in several other commonly used structures, the zero-IF structure receiver has the following advantages. First, there is no image interference and the receiver therefore does not need an image suppression filter or an IF filter, so the cost of the receiver is low and the integration is high. Second, the final output signal of the receiver is a baseband signal, so the gain control and channel selection can be carried out in the baseband, and the circuit design complexity is greatly reduced. Third, there is no mirror image suppression in the receiver, so the problem of matching the filter and low-noise amplifier (LNA) can be basically ignored.

The transceiver antenna is responsible for the electromagnetic wave transmitting and receiving devices and has energy conversion and directional functions. In this article, the antenna includes a waveguide slot antenna, which can effectively reduce the volume (especially the thickness) compared with the reflector antenna, reduce the production and debugging difficulties, and help control the quality of the product. The antenna pattern is the change graph of the antenna radiation characteristic with the spatial coordinates, comprising an important antenna indicator. In the antenna design process, to ensure a high azimuthal aperture synthesis quality, the horizontal beamwidth of the antenna pattern was 90°, and to ensure a favorable antenna gain, the vertical beamwidth was 4°. Due to the reciprocity of antenna transmission and reception, the parameters of the receiving antenna pattern are basically the same as those of the transmitting antenna pattern. Additionally, to reduce signal leakage and coupling between antennas and improve antenna transmission and reception isolation, the antenna sidelobes should be suppressed as much as possible, the distance between the transmitting and receiving antennas should be widened, and partitions should be installed between the transmitting and receiving antennas.

The RTAAP module first collects zero-IF signals through an analog-to-digital converter (ADC) with a high quantization number; then it performs real-time signal processing based on the embedded graphics processing unit (GPU) architecture.

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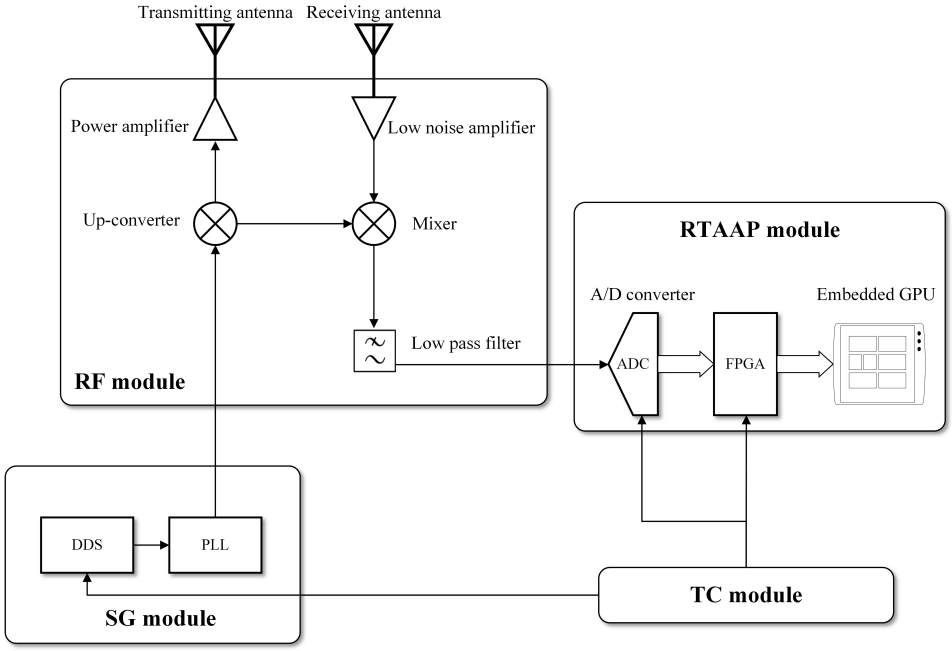


Fig. 2. Architecture diagram of the radar sensor.

TABLE I
RADAR WORKING PARAMETERS

Parameters	Values
Carrier frequency/GHz	94
Bandwidth/GHz	1
Pulse repetition frequency/Hz	300
Sampling frequency/MHz	20
Sampling quantization bits/bits	16
Pulse width/ms	2
Antenna beam angle (vertical)/deg	4
Antenna beam angle (horizontal)/deg	90
Rotating arm length/m	1

Finally, the module outputs the imaging and detection results through the network.

The TC module mainly generates the system working timing and includes various working clocks. The module coordinates the functionalities between the RF and RTAAP modules and the turntable through the crystal oscillator, internal logic, and external interfaces.

3) Radar Working Parameters: Table I shows the working parameters of the radar system. When the system turntable speed is 360°/min, the step size of each pulse is 0.02°, which corresponds to a distance of 0.35 mm. This distance is much smaller than the 3.2-mm wavelength that corresponds to the center frequency, so the azimuth resolution of the system is as follows:

$$\Delta a = \arcsin\left(\frac{c}{4 f_c L \sin\left(\frac{\emptyset_h}{2}\right)}\right) = \arcsin\left(\frac{3e8}{4 \times 94e9 \times 1 \times \sin\left(\frac{45}{180} \times \pi\right)}\right) = 0.06^\circ \quad (2)$$

where L is the length of the rotating arm, f_c is the carrier frequency, and $\emptyset_h$ is the antenna horizontal beam angle.

III. INFORMATION PROCESSING FLOW

The information processing section first produces the radar transceiver signal model, and then analyzes the imaging geometric model. Next, it presents the 2-D frequency-domain algorithm and imaging processing flow, and then provides the FOD detection algorithm and flow.

A. Radar Transceiver Signals

This system uses a linear FMCW signal [31], and its time-domain diagram is shown in Fig. 3, where T_p is the repetition period. The radar transmitting signal is as follows:

$$s_T(t) = \text{rect}\left(\frac{t}{T_{\text{pu}}}\right) \exp\left[j2\pi\left(f_c t + \frac{1}{2}k_r t^2\right)\right], \quad t \in [0, T_{\text{pu}}] \quad (3)$$

where T_{pu} is the sweep duration, k_r is the chirp constant, and t is the time within a modulation period. The microwave is reflected by the target, and then the received signal from the radar is

$$s_R(t) = A s_T(t - \Delta t) \quad (4)$$

where Δt is the echo delay, $\Delta t = 2R/c$. R is the distance between the target and the radar, and A is the amplitude attenuation coefficient.

The system receiver mixes the received signal with the reference to obtain the demodulated IF signal

$$s_{\text{IF}}(t) = A s_T(t - \Delta t) s_T^*(t) = A \exp\left[-j2\pi\left(f_c \Delta t + k_r \Delta t \cdot t - \frac{1}{2}k_r \Delta t^2\right)\right] \quad (5)$$

where $t \in (\Delta t, T_{\text{pu}})$.

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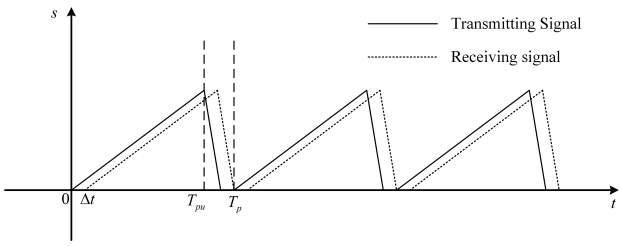


Fig. 3. Time-domain diagram of the linear FMCW signal.

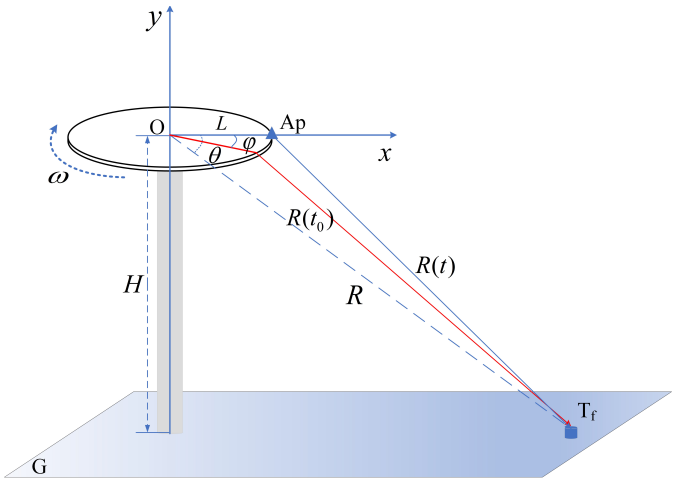


Fig. 4. Geometric model of AS-SAR operation.

The fast Fourier transform (FFT) of the IF signal yields

$$\begin{aligned} S_{\mathrm{IF}}(f) &= \int_{\Delta t}^{T_{\mathrm{pu}}} s_{\mathrm{IF}}(t) e^{-j2\pi f t} dt \\ &= A_1 \sin c\left[\pi (f - k_r \Delta t) T_{\mathrm{pu}}\right] \\ &\quad \cdot e^{j\pi k_r \Delta t^2} e^{-j2\pi f_c \Delta t} e^{-j2\pi f \Delta t} \end{aligned} \tag{6}$$

where $A_1 = Ak_r/2$, and pulse compression is related to the process of solving $S_{\mathrm{IF}}(f)$. By analyzing $S_{\mathrm{IF}}(f)$, we can see that its peak position is

$$f = k_r \Delta t = k_r \frac{2R(t)}{c}. \tag{7}$$

The IF signal frequency can be used to map the target distance information. For targets at different distances, the return time Δt is also different. According to the formula, there is no near-end blind zone in the radar, and the distance is 0, corresponding to $S_{\mathrm{IF}}(f)$ frequency of 0. Upon combining the characteristics of AS-SAR omnidirectional scanning methods, the radar system can achieve "blind-zone-free" detection.

B. Image Processing

The AS-SAR antenna is placed on a fixed-length mechanical rotating arm, where its rotation can be used to achieve relative movement with the target and then obtain Doppler information. The geometric model of AS-SAR operation is given in Fig. 4, where G is the surface plane; O is the center of the rotary axis, with the FOD target T_f placed on the G plane; H is the radar height; Ap is the phase center of the radar antenna; L is the length of the rotating arm (the distance from Ap to the rotary axis); ω is the angular velocity of the turntable, with the circular angle $\varphi = \omega(t - t_0)$ corresponding to the radar antenna Ap at time t where t_0 is the time corresponding to the closest distance between the antenna Ap and the target; and θ is the angle between the OT_f line and the rotation plane of the radar rotating arm. Suppose the distance from the center of the rotation axis O to the target T_f is R and the distance from the antenna Ap to the target at time t is $R(t)$. Then, from the geometric AS-SAR model, the expression of $R(t)$ is obtained as

$$\begin{aligned} R(t) &= \sqrt{R^2 + L^2 - 2RL \cdot \cos\theta \cdot \cos\omega(t - t_0)} \\ &= \sqrt{1 + (L/R)^2 - 2(L/R) - 2RL \cdot \cos\theta \cdot \cos\omega(t - t_0)}. \end{aligned} \tag{8}$$

In the actual radar scanning imaging process, because $R \gg L$, the above equation can be approximated through Taylor expansion as

$$R(t) \approx R - L \cdot \cos\theta \cdot \cos\omega(t - t_0). \tag{9}$$

To achieve the goal of improving the azimuth resolution of the radar, the radar sensor is placed on the turntable, and an arc synthetic aperture is formed in space through a rotating motion. The imaging process focuses on and coherently accumulates the echo signals within the radar accumulation angle. To analyze the imaging algorithm, substituting $\Delta t = 2R(t)/c$ into $s_{\mathrm{IF}}(t)$ formula yields

$$\begin{aligned} &s_{\mathrm{IF}}(t) \\ &= A \exp\left[-j2\pi\left(k_r \frac{2R(t)}{c} \cdot t + f_c \frac{2R(t)}{c} - \frac{1}{2} k_r \left(\frac{2R(t)}{c}\right)^2\right)\right]. \end{aligned} \tag{10}$$

Then, the corresponding $S_{\mathrm{IF}}(f)$ is

$$\begin{aligned} S_{\mathrm{IF}}(f) &= A_1 \cdot \sin c\left[\pi T_{\mathrm{pu}}\left(f - k_r \frac{2R(t)}{c}\right)\right] \\ &\quad \cdot e^{-j4\pi f \frac{R(t)}{c}} e^{j4\pi k_r \frac{R(t)}{c^2}^2} e^{-j4\pi f_c \frac{R(t)}{c}}. \end{aligned} \tag{11}$$

There are three exponential terms corresponding to the distance phase, residual video phase (RVP), and azimuthal Doppler phase. Therefore, the process of AS-SAR imaging can be divided into three steps. One step is the distance-dimensional pulse compression, which performs matched filtering in fast time, achieves deskew processing, and improves the signal-to-noise ratio (SNR) of the radar; the pulse compression process of an FM continuous regime radar is deskewed processing and FFT operation. The second step is the RVP compensation operation, which aligns the echo in the desired range. The third step is the azimuth-dimensional compression, which can be realized by time- or frequency-domain algorithms.

Currently, AS-SAR imaging algorithms mainly include time-domain correlation, back-projection (BP) [32], frequency-domain range–Dopple (RD) [33], [34], and chirp scaling (CS) [35] methods. The time-domain algorithm

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obtains high-resolution SAR images through slow-time dimension "phase compensation-coherent accumulation" operation, which is computationally intensive but not limited by the form and motion of the radar arrays and is therefore widely used [32], [36], [37]. The frequency-domain imaging algorithm has the advantages of simple principles, easy implementation, and high operational efficiency and is widely used in real-time processing. Taking the real-time processing requirements of the AS-SAR system into account, the 2-D frequency-domain algorithm is selected for research and analysis here.

According to the "stop-and-go" assumption in SAR imaging, the 1-D pulse compression corresponds to the fast time, the azimuth φ corresponds to the slow time, and the system forms the accumulated aperture with the azimuth, ignoring the change in distance between the radar and the target caused by the difference time between signal transmission and reception. According to the expression of RVP in (11), the change in RVP caused by arc-scanning aperture is less than 1.4×10^{-4} radians (rad). So RVP has little impact on imaging quality and its value can be ignored. Then $S_{\mathrm{IF}}(f)$ is written as 2-D

$$S(f,\varphi) = A_1 \sin c\left[\pi T_{\mathrm{pu}}\left(f - k_r \frac{2(R - L\cdot\cos\theta\cdot\cos\varphi)}{c}\right)\right] \cdot e^{-j4\pi f \frac{R-L\cdot\cos\theta\cdot\cos\varphi}{c}} e^{-j4\pi f_c \frac{(R-L\cdot\cos\theta\cdot\cos\varphi)}{c}}. \tag{12}$$

When the target is closest to the radar antenna, $\varphi = 0$. Considering the fact that the distance between the target and the radar axis is constant and there are different delays only for different azimuth angles φ, the Fourier transform of $S(f,\varphi)$ along the azimuth φ is expressed as

$$S(f, f_\varphi) = \int S(f,\varphi)\cdot e^{-j2\pi f_\varphi\cdot\varphi}\cdot d\varphi \tag{13}$$

where f_φ is the frequency of the rotating arm circular angle φ. Assume that the horizontal beamwidth of the antenna is φ_w, whose band is related to the rotational speed of the rotary table ω and the carrier frequency wavelength λ, which can be calculated with the following equation: the bandwidth of f_φ is related to the rotation speed of the turntable ω and the carrier frequency wavelength λ, which can be calculated with the following equation:

$$B_\varphi = \frac{4L\omega\sin\left(\frac{\varphi_w}{2}\right)\cos\theta}{\lambda}. \tag{14}$$

$S(f, f_\varphi)$ can also be expressed as

$$S(f, f_\varphi) = \int_{-\frac{\varphi_w}{2}}^{\frac{\varphi_w}{2}} A_1 \cdot \sin c\left[\pi T_{\mathrm{pu}}\left(f - k_r \frac{2(R - L\cdot\cos\theta\cdot\cos\varphi)}{c}\right)\right] \times \mathrm{rect}\left(\frac{\varphi}{\varphi_w}\right)\cdot e^{-j4\pi f \frac{R-L\cdot\cos\theta\cdot\cos\varphi}{c}} e^{-j4\pi f_c \frac{(R-L\cdot\cos\theta\cdot\cos\varphi)}{c}} \cdot e^{-j2\pi f_\varphi\cdot\varphi} d\varphi. \tag{15}$$

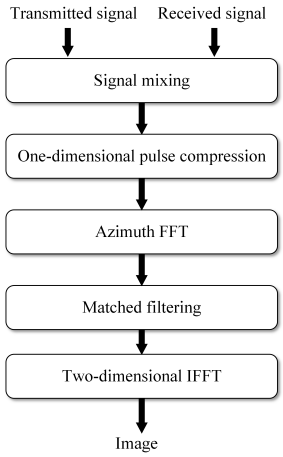


Fig. 5. Imaging processing flow of the radar.

To simplify the calculation, let the target be in the far-field region and in the antenna rotation plane; then $\theta = 0$. Moreover

$$\begin{aligned} P(\varphi) &= -j4\pi f \frac{R - L\cdot\cos\theta\cdot\cos\varphi}{c} - j4\pi f_c \frac{(R - L\cdot\cos\theta\cdot\cos\varphi)}{c} - j2\pi f_\varphi\cdot\varphi \\ &= -j4\pi (f + f_c)\frac{R - L\cdot\cos\varphi}{c} - j2\pi f_\varphi\cdot\varphi. \end{aligned} \tag{16}$$

According to the stationary phase principle, let $(dP(\varphi)/d\varphi) = 0$, then

$$f_\varphi = -\frac{2(f + f_c)}{c} L\sin\varphi. \tag{17}$$

Therefore, the relationship between φ, $R(t)$ and f_φ can be obtained as follows:

$$\varphi = \varphi(f_\varphi) = -\arcsin\left[\frac{c}{2L(f + f_c)} f_\varphi\right] \tag{18}$$

$$\begin{aligned} R(t) = R(f_\varphi) &\approx R - L\cdot\cos\left(-\arcsin\left(\frac{c}{2L(f + f_c)} f_\varphi\right)\right) \\ &\approx R - \sqrt{L^2 - \left(\frac{cf_\varphi}{2(f + f_c)}\right)^2}. \end{aligned} \tag{19}$$

Considering that the spectrum of the same target with different azimuth angles in the frequency domain differs only in the linear phase, imaging is the process of compensating for the phase. Therefore, without affecting the phase analysis, let the amplitude of $S(f, f_\varphi)$ be $A_2(f, f_\varphi)$; then

$$S(f, f_\varphi) = A_2(f, f_\varphi) \cdot \exp\left[-j4\pi (f + f_c)\frac{R(f_\varphi)}{c} - j2\pi f_\varphi\cdot\varphi(f_\varphi)\right]. \tag{20}$$

Phase-matching filters can be designed

$$M(f, f_\varphi) = \exp\left[j4\pi (f + f_c)\frac{R(f_\varphi) - R}{c} + j2\pi f_\varphi\cdot\varphi(f_\varphi)\right]. \tag{21}$$

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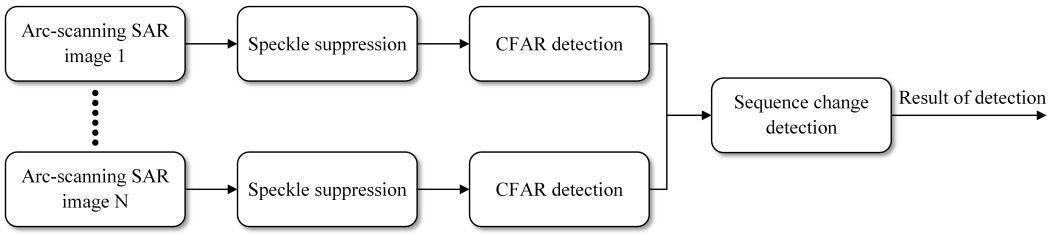


Fig. 6. AS-SAR detection processing flow.

Then, the matched filter output is

$$S_M(f, f_\varphi) = S(f, f_\varphi) \cdot M(f, f_\varphi) = A_2(f f_\varphi)\exp\left[-j4\pi (f + f_c)\frac{R}{c}\right]. \quad (22)$$

Finally, the 2-D inverse fast Fourier transform (IFFT) is performed on $S_M(f, f_\varphi)$, and the time-domain SAR image $s_I(R, \varphi)$ in polar coordinates can be obtained.

In summary, the AS-SAR imaging processing flow in this article includes signal acquisition, signal mixing, 1-D range pulse compression, azimuth FFT, matched filtering, 2-D IFFT, etc. The flowchart is given in Fig. 5.

C. Detection Processing

Detection is an important aspect in the automatic extraction of FOD targets. The main challenges are as follows. First, FOD on the runway is caused by many factors, such as a piece of asphalt concrete or cement concrete, a runway lighting part, a wrench, a sleeve, a twisted metal strip, or the fuel tank cap of an aircraft or vehicle; this means that the targets present different shapes in the AS-SAR image. Second, FOD targets are small in size and weak in scattering intensity, which present as weak targets in the images. Third, the people, vehicles, and aircraft on runways may form strong scattering points in the images, affecting the target detection performance.

This system can monitor the airport runway and other areas for a long time and form sequential AS-SAR images. Therefore, sequence change detection can be performed in response to the abovementioned detection challenges. The main process is shown in Fig. 6, which mainly includes speckle suppression, constant false alarm rate (CFAR) detection, and sequence change detection.

1) Speckle Suppression: The speckle in the AS-SAR image is formed by the coherent accumulation of multiple scattering echoes received by each resolution unit, seriously affecting the image interpretation. Therefore, for sequential AS-SAR images, speckle suppression is carried out first. Typical speckle suppression methods include the spatial-domain filtering algorithm, frequency-domain filtering algorithm, multiview processing algorithm, and nonlocal-mean filtering algorithm. These algorithms have their own advantages and disadvantages for different application scenarios. Considering the fact that the monitoring object of the system is the runway and the background image has strong stationary characteristics, the Lee filter in the spatial-domain filtering algorithm can be used. The Lee filter [38] assumes that the SAR image satisfies multiplicative noise

$$s_I(R, \varphi) = s_u(R, \varphi) \cdot n \quad (23)$$

where $s_u(R, \varphi)$ is the radar scattering coefficient and n is the speckle noise. Considering the stability of the image in the monitoring area, parameters such as mean and variance can be approximated as the sample mean and variance within the filtered sliding window. Then, the filtering method is adopted

$$\hat{s}_u(R, \varphi) = \bar{s}_I(R, \varphi) + \frac{\sigma_{s_v}^2 - \bar{s}_I(R, \varphi)^2 \cdot \sigma_n^2}{\sigma_{s_v}^2\left(1 + \sigma_n^2\right)} \times (s_I(R, \varphi) - \bar{s}_I(R, \varphi)) \quad (24)$$

where $\bar{s}_I(R, \varphi)$ is the average pixel value in the sliding window, $\sigma_{s_v}^2$ is the pixel variance in the sliding window, and σ_n^2 is the image noise variance. The Lee filter considers the statistical properties of noise and uses local statistical information for minimum mean-square-error estimation, so it can be used to achieve better speckle suppression in stationary high-resolution AS-SAR images to better retain target edge features.

2) CFAR Detection: In practical situations, it is difficult to detect targets using fixed thresholds due to different target locations, where the thresholds need to be determined adaptively. CFAR detector is widely used for SAR target detection, and it involves a kind of algorithm that exploits the difference between target grayscale and clutter grayscale of its surrounding neighborhood for separation [39]–[41]. The CFAR detection process is shown in Fig. 7. The constant false alarm probability (P_f) and image distribution are the two key factors. P_f is generally a constant, which is the expected CFAR value before detection. The image distribution describes the statistical characteristics of the cluttered background.

The detailed process of the CFAR detector is as follows.

1) Calculate the image distribution, setting P_f of detection.
2) Traverse each pixel in the image, delineate neighboring pixels according to the sizes of the target area, protected area, and background area in the sliding window, as well as other criteria (e.g., selection of large criterion, selection of small criterion), and use this information to estimate the statistical parameters.
3) Calculate the local threshold T_l based on these parameters.
4) Compare this test pixel value with T_l and set it to zero if it is less than the threshold value; keep it if it is greater than the threshold value.

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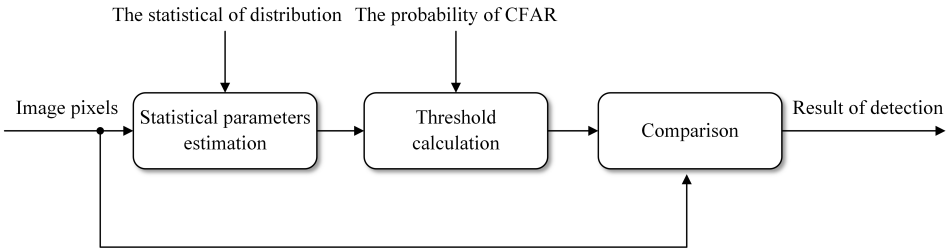


Fig. 7. CFAR detection flow.

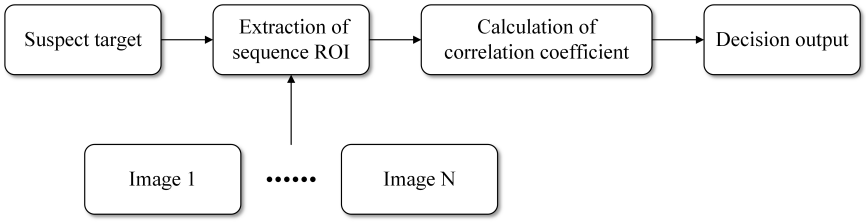


Fig. 8. Sequence change detection flow.

5) Determine whether the entire image has been processed; if not, move to the next pixel and return to Step 2.

3) Sequence Change Detection: With FOD target detection, it is necessary to ensure a high detection rate to improve the detection performance of the system and prevent missed alarms; additionally, it is necessary to ensure a low false alarm rate to improve the practicality of the system and prevent a situation involving continuous research and judgment of false targets. Therefore, it is necessary to make full use of AS-SAR system capability to continuously image the runway area and process the sequence images through change detection to obtain better processing results. In this article, a method of tracking after detection is used for sequence change detection, where the similarity is determined by calculating the coherence coefficients of pixels in the same area. When the suspected target suddenly appears, the coherence coefficient in the region of interest (ROI) increases, while the coherence coefficient of the nontarget region is lower.

The sequence change detection process is given in Fig. 8. After CFAR detection, the corresponding ROI of the suspected target is extracted, and then the correlation coefficient of the sequence is calculated with the formula

$$c_{n-1} = \frac{\left|\sum_{i=1}^{I}\sum_{j=1}^{J} x_n(i,j)x_{n-1}^*(i,j)\right|}{\sqrt{\sum_{i=1}^{I}\sum_{j=1}^{J}|x_n(i,j)|^2\sum_{i=1}^{I}\sum_{j=1}^{J}|x_{n-1}(i,j)|^2}} \tag{25}$$

where I and J are the range and azimuth window size of the ROI area respectively, n is the number of sequence frames, and $*$ denotes the complex conjugate. Then a sequence change detection is performed for N frame images, where $N-1$ correlation coefficients are obtained; these coefficients are represented by the vector $\vec{F}_c = [c_1 c_2, \ldots, c_{N-1}]$. Finally, through existing sample training, according to $\vec{F}_c$ change law, a threshold is established and a judgment is conducted to determine whether the suspected target is an FOD target. It is worth noting that in addition to the correlation coefficient feature, target identification can also be performed by feature selection to remove false alarms [42]–[44].

IV. Flicker Clutter Suppression Simulation

Reducing false alarms while ensuring a high detection rate is an important challenge for FOD detection systems. Because the AS-SAR imaging algorithm uses virtual aperture coherent accumulation, the system has a good suppression effect on the flicker clutter that forms on the runway during severe weather such as rainfall; this good suppression effect is verified by related simulation experiments.

Raindrops fall through the air and are affected by gravity, air resistance, etc. As their speed increases, large raindrops split to form smaller raindrops, or small raindrops collide with other small raindrops to merge together and form large water droplets. Therefore, the raindrop spectrum is nonuniform. Affected by the type of rainfall, region, and climatic conditions, the raindrop spectrum has different forms. In general, raindrops are small in size, with a diameter of less than 4 mm, and their durations in the air are short, so they do not appear directly in the AS-SAR image but induce only microwave attenuation. When the rainfall is heavy, a water film is formed on the ground. When raindrops fall, they can impact and produce a crown-shaped cylinder, as shown in Fig. 9. The crown-shaped cylinder radius and height are comparable in size to the FOD target, and the duration of this shape can reach 60 ms. Moreover, this shape size first increases and then decreases, manifesting as flicker clutter. Therefore, the simulation experiment here mainly aims to study this phenomenon.

According to previous research [45]–[48], the ratio of the raindrop crown height to the raindrop diameter is η_c

$$\eta_c = \frac{H}{D_0}. \tag{26}$$

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Fig. 9. Raindrop crown model.

This value is related to the Weber number W. When $W = 296$, η_c is close to 6, where D_0 is the diameter of the raindrop, and the height of the raindrop crown is 2.4 cm. The formula for the diameter of the raindrop crown is

$$\Delta_c = C(\tau - \tau_0)^n \tag{27}$$

where C is a constant coefficient (its value is 1.5 when the Weber number is 296), the empirical value of n is 0.43 ± 0.03, and τ_0 is the time at which the raindrop impacts the water film. Assuming the maximum value of the crown occurs at 20 ms, the diameter is 5.4 cm.

Rainfall observations can intuitively indicate rainfall intensities, which can be divided into light rain, moderate rain, heavy rain, large heavy rain, etc. Considering the fact that the formation of surface water films generally requires intensities above moderate rain and that rainfall intensities above large heavy rain events are unfavorable for aircraft landing and takeoff, this part of the simulation is carried out for moderate rain and heavy rain. According to regulations, the 24-h rainfall range for moderate rain is 10–24.9 mm, and the 24-h rainfall range for heavy rain is 25–49.9 mm. According to (26), the crown formed by heavy raindrops is larger and induces a greater impact on detection. Therefore, assuming that the size of the raindrops is $D_0 = 4$ mm, the raindrop volume is

$$V = \frac{4}{3}\pi (D_0/2)^3 = 3.35 \times 10^{-8}\ \mathrm{m}^3. \tag{28}$$

Additionally, taking the rainfall of moderate rain and heavy rain as 24.9 and 49.9 mm, respectively, the number of raindrops per square meter per second can be calculated as 10.36 and 20.72, respectively.

This article simulates and compares the imaging results of AS-SAR and arc-scanning real aperture radar (AS-RAR) under different rainfall conditions. The imaging angle ranges from 45° to 135°, the imaging distance ranges from 25 to 35 m, and the radar scanning speed is 3.3 revolutions per minute (r/min). Assuming that there is an FOD target with a height of 2.4 cm and a diameter of 5.4 in the center of the monitoring area, the scattering characteristics are the same as those of the raindrop crown. The AS-SAR antenna beamwidth is 90°, and the AS-RAR antenna beamwidth is 4°, the same as Xsight's FODetect. Fig. 10 displays a comparison between AS-SAR and AS-RAR imaging results. The radar images of the FOD targets are shown in Fig. 10(a) and (b). Clearly, the azimuth resolution of the AS-SAR image is higher than that of the AS-RAR image.

TABLE II
INTENSITY OF THE BACKGROUND CLUTTER IN AN AS-SAR IMAGE

Rainfall intensity	Mean value of the strongest clutter/dB
Moderate rain	-35.89
Heavy rain	- 22.98

Raindrops fall randomly and uniformly in the whole scanning area. In the polar coordinate simulation, the raindrops are uniformly distributed in the azimuth direction and uniformly distributed in the squaredistance range. The imaging results under rainy conditions are shown in Fig. 10(c)–(f). Fig. 10(c) and (d) shows the imaging results of FOD targets in moderate rain. In Fig. 10(c), the FOD target intensity is significantly higher than that of the background clutter. However, in Fig. 10(d), the FOD target has the same intensity level as the background clutter. In moderate rainfall conditions, the detection performance of AS-RAR is seriously reduced. Moreover, under heavy rainy conditions, the situation worsens. As shown in Fig. 10(f), the FOD target is completely submerged in the background clutter. In contrast, the intensity of the FOD target in the AS-SAR image is still stronger than that of the background clutter, as shown in Fig. 10(f).

The larger the raindrops, the stronger the background clutter. To quantitatively and accurately analyze the background clutter caused by rainfall, Table II includes the intensity of the background clutter in the AS-SAR image. For 300 simulation tests, the mean value of the strongest clutter is -35.89 dB for moderate rain and -22.98 dB for heavy rain; these results are consistent with the situations shown in Fig. 10(c) and (e).

For AS-SAR, it can be observed that under heavy rainfall conditions, the target intensity is still much higher than the background, achieving effective FOD detection; likewise, the system can suppress flicker clutter such as rainfall. However, AS-RAR has a limited ability to suppress flicker clutter, and the detection performance severely deteriorates with the application of rainfall conditions.

V. EXPERIMENTAL RESULTS

Based on the description of the experimental scene, this section processes and analyzes the measured data on sunny and rainy days to study the imaging and detection results of the radar on the FOD of small targets under two weather conditions. For ease of description, we abbreviate AS-SAR-based FOD detection system as AS-FODR.

A. Introduction and AS-FODR Imaging Results for the Experimental Scene

A photograph of the AS-FODR designed in this article is presented in Fig. 11, where the radar antenna base, rotating arm, and base are externally visible. To verify the detection performance of this system, the system is installed at the general airport in Dalu, Zhenjiang, Jiangsu province. Fig. 12 shows a satellite photograph of the experimental scene. The fan shape indicates the radar monitoring area, and the black dot indicates the location of the radar. An optical

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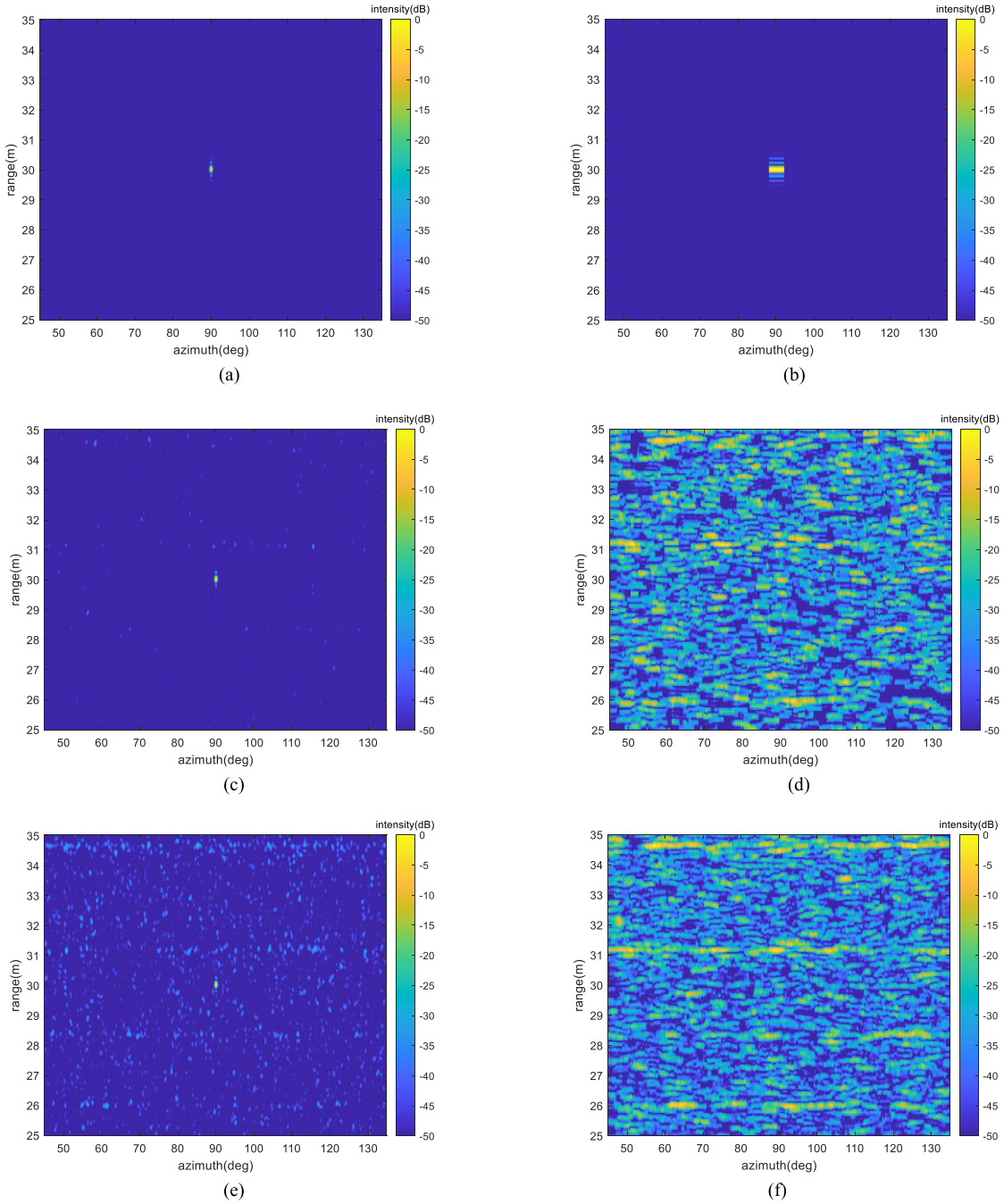


Fig. 10. Comparison of the AS-SAR and AS-RAR imaging results: (a) AS-SAR imaging results for FOD; (b) AS-RAR imaging results for FOD; (c) AS-SAR imaging results in moderate rain; (d) AS-RAR imaging results in moderate rain; (e) AS-SAR imaging results in heavy rain; and (f) AS-RAR imaging results in heavy rain.

photograph of the runway is shown in Fig. 13, where the runway is also marked with black arrows. Fig. 13(a) shows the scene when it rains, while Fig. 13(b) shows the scene when it is sunny.

Fig. 14 shows the AS-FODR imaging result for the experimental scene, as expressed in a Cartesian coordinate system. The radar position is the origin of the coordinate axis, the horizontal axis represents the azimuth, and the vertical axis represents the distance. The runway is marked with a black arrow. The AS-FODR can achieve high-resolution imaging of large areas and can effectively monitor runways. Similar to the previous method, the runways are marked with white arrows.

To study the small target detection ability of AS-FODR, metal cylinders are used as FOD targets. The heights of each target were equal their diameters, and the diameters were 1, 1.5, 2, 3, and 4 cm, as shown in Fig. 15. During the text, these cylindrical targets were placed on the runway, which was more than 250 m away from the radar, on the left side of the apron to the runway connection.

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Fig. 11. AS-FODR.

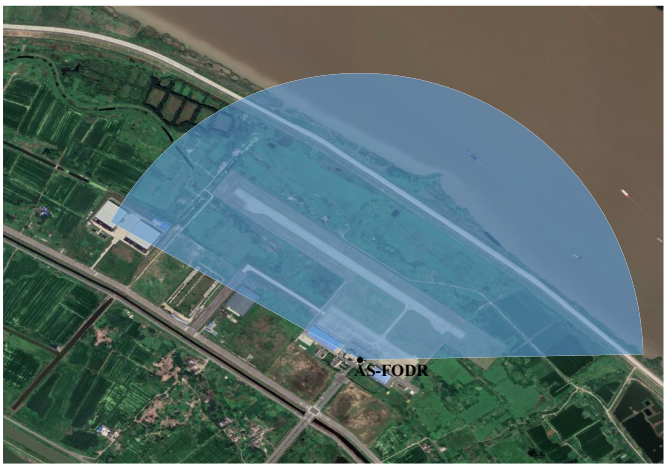


Fig. 12. Satellite photograph of the experimental scene.

B. AS-FODR Imaging and Detection Results on a Sunny Day

On April 19, 2021, we carried out a field detection test of the FOD target on a sunny day. Fig. 16 shows the AS-FODR image of the runway area after the targets are placed. The targets more clearly appear on the left runway in Fig. 16(a). The five types of targets with diameters ranging from 1 to 4 cm are separated by 5 m. The targets are arranged in order from weak to strong in the image, which is consistent with the actual deployment scene. Fig. 16(b) gives the enlarged image of white rectangular area, and Fig. 16(c) shows the corresponding detection results. For comparative analysis, the parameters of the CFAR detector except the false alarm rate are fixed. We can change the threshold of the detector by adjusting the false alarm rate. When there is no false alarm in the detected image, we can not only know whether the target can be detected effectively but also compare the image quality under different conditions through the target size and intensity. According to Fig. 16(c), the metal cylinders with diameters ranging from 1 to 4 cm in the white rectangular area can be extracted effectively. In summary, the AS-FODR can achieve long-range FOD small target detection on sunny days.

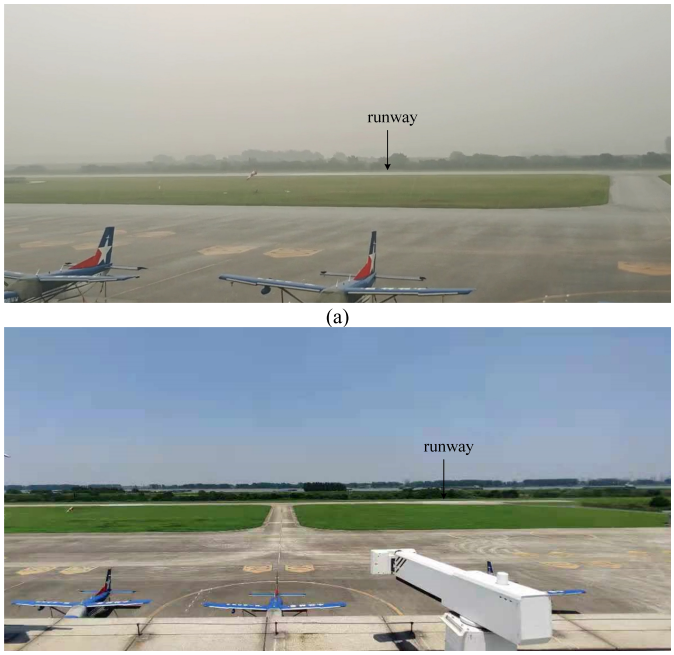


Fig. 13. Optical photograph of the experimental scene: (a) optical photograph of the runway on a rainy day and (b) optical photograph of the runway on a sunny day.

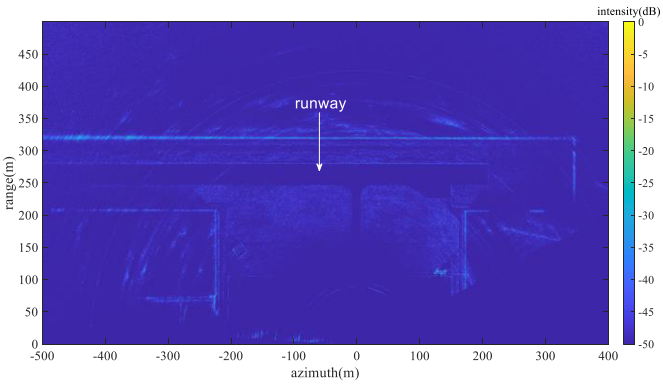


Fig. 14. AS-FODR imaging result for the experimental scene.

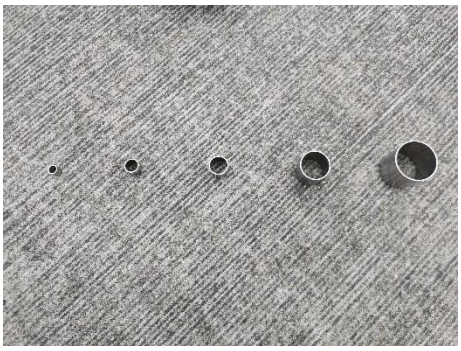


Fig. 15. Photograph of the cylindrical targets.

C. AS-FODR Imaging and Detection Results on a Rainy Day

On May 15, 2021, we carried out a field detection test of the FOD target on a rainy day. Fig. 17(a) displays the AS-FODR image of the runway area after the targets are placed, and Fig. 16(b) gives the enlarged image of white rectangular area. Compared with Fig. 16(a) and (b), the noise of the runway is

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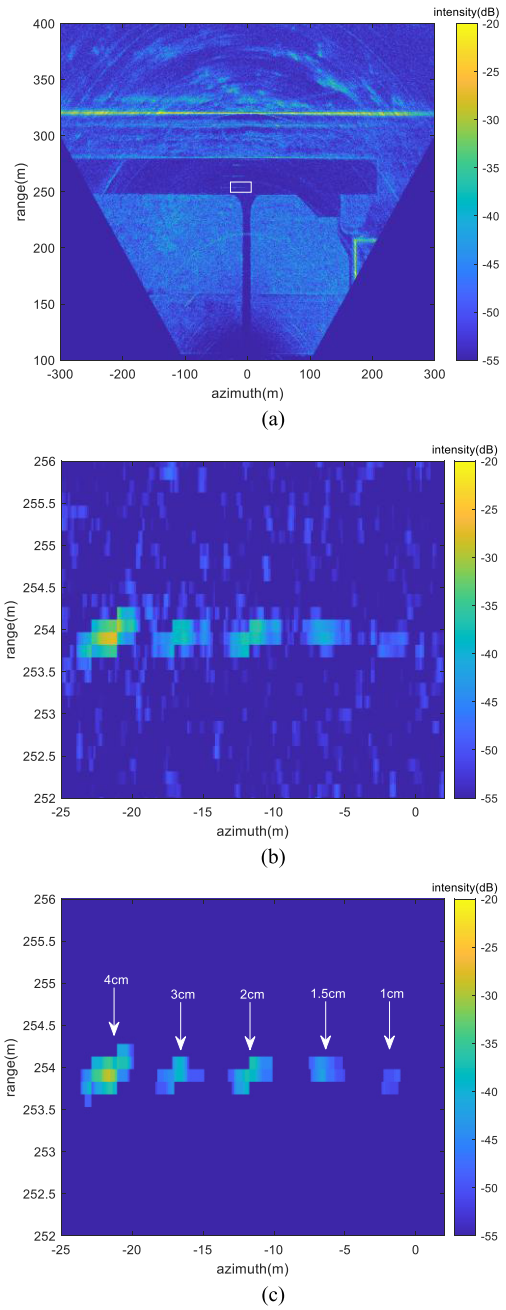


Fig. 16. AS-FODR image of the cylindrical targets on a sunny day: (a) AS-FODR image of the runway; (b) enlarged image of white rectangular area; and (c) cylinder detection results.

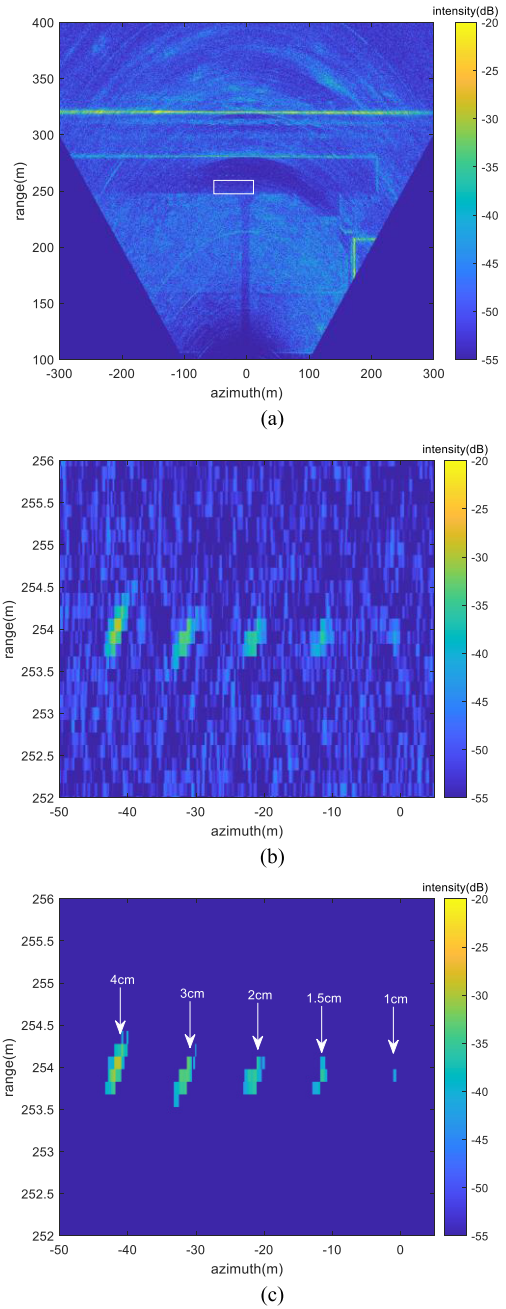


Fig. 17. AS-FODR image of the cylindrical targets on a rainy day: (a) AS-FODR image of the runway; (b) enlarged image of white rectangular area; and (c) cylinder detection results.

significantly increased on a rainy day; however, the targets are still visible on the left side of the runway. The five types of targets with diameters ranging from 1 to 4 cm are separated by 10 m. The targets are arranged in order from weak to strong in the image, which is consistent with the actual deployment scene. The corresponding detection results are given in Fig. 17(c). Cylinders with diameters ranging from 1 to 4 cm in the white rectangular area can be extracted effectively, but the shape size of the FOD targets extracted on a rainy day is smaller than those shown in the detection results in Fig. 16(c) based on a sunny day. In summary, it can be concluded that the AS-FODR can achieve the detection of small FOD targets over a long range on rainy days.

D. Comparing the Results Between Synthetic Aperture and Real Aperture Data

To further demonstrate the advantages of AS-FODR through the measured data, Fig. 18 displays the images of synthetic aperture and real aperture (without azimuth compression) in a polar coordinate system. Fig. 18(a) and (b) data were obtained on March 27, 2021, which was a sunny day. During the experiment, a metal cylinder with a diameter of 1.5 cm was placed on the runway, and another metal cylinder with a diameter of 1 cm was placed on the vertical connecting road. The synthetic aperture image of the cylindrical targets is shown in Fig. 18(a). They are clearly visible with high range and azimuth resolution, and their intensity is also higher than that

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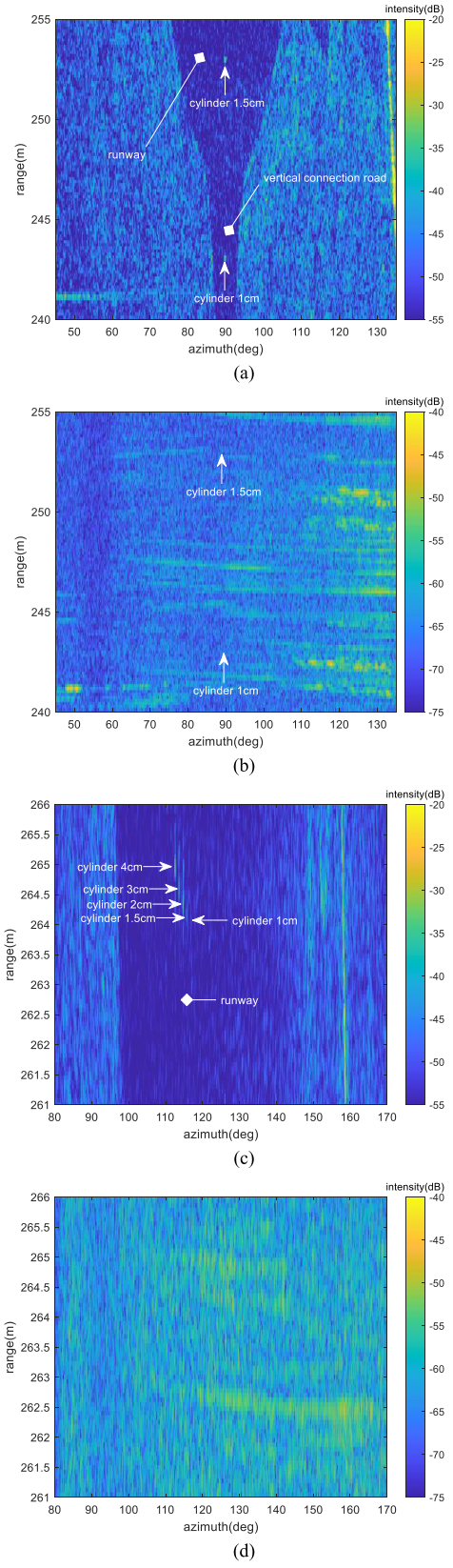


Fig. 18. Images of synthetic aperture and real aperture: (a) synthetic aperture image on a sunny day; (b) real aperture image on a sunny day; (c) synthetic aperture image on a rainy day; and (d) real aperture image on a rainy day.

of the neighborhood background clutter. Fig. 18(b) is a real aperture image, where the cylindrical targets are faintly visible,

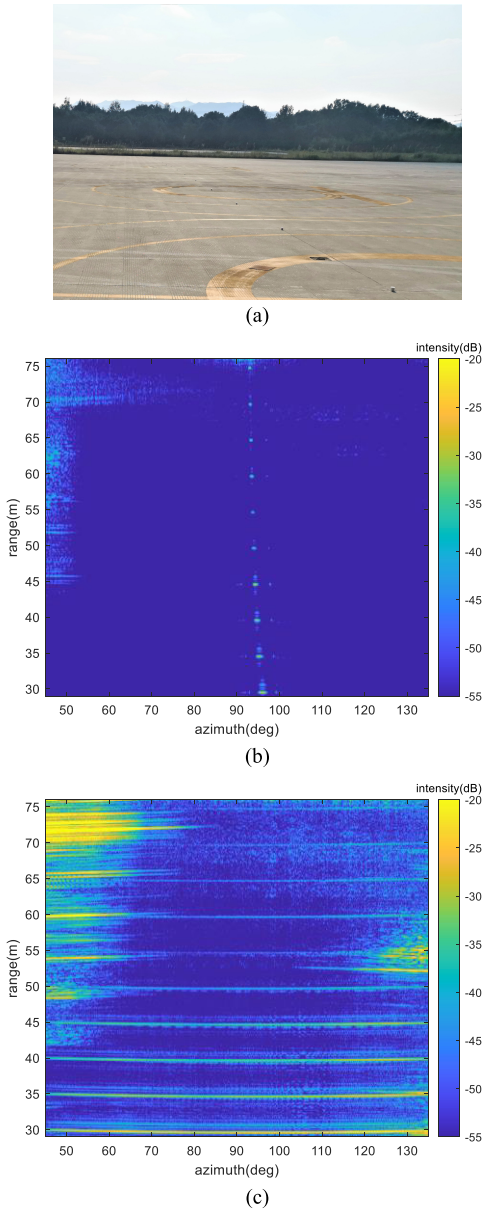


Fig. 19. Experimental test result of azimuth compression: (a) optical photograph of the experimental scene at Kaihui general airport in Changsha, Hunan province; (b) synthetic aperture image; and (c) real aperture image.

and their intensity is comparable to that of the background clutter.

Fig. 18(c) and (d) data were obtained on October 15, 2021, which was a rainy day. Metal cylinders with diameters ranging from 1 to 4 cm are separated by 5 m on the runway. In the synthetic aperture image shown in Fig. 18(c), although the noise has increased, the cylindrical targets are still visible. However, in the real aperture image shown in Fig. 18(d), the clutter intensity of the runway is greatly increased, and the cylindrical targets are submerged in the background clutter.

According to Fig. 18, the intensity and SNR of the target are greatly improved after azimuth compression. Synthetic aperture processing can effectively suppress flicker clutter such as rainfall.

To further clearly demonstrate the impact of azimuth compression on target azimuth resolution, we conducted an

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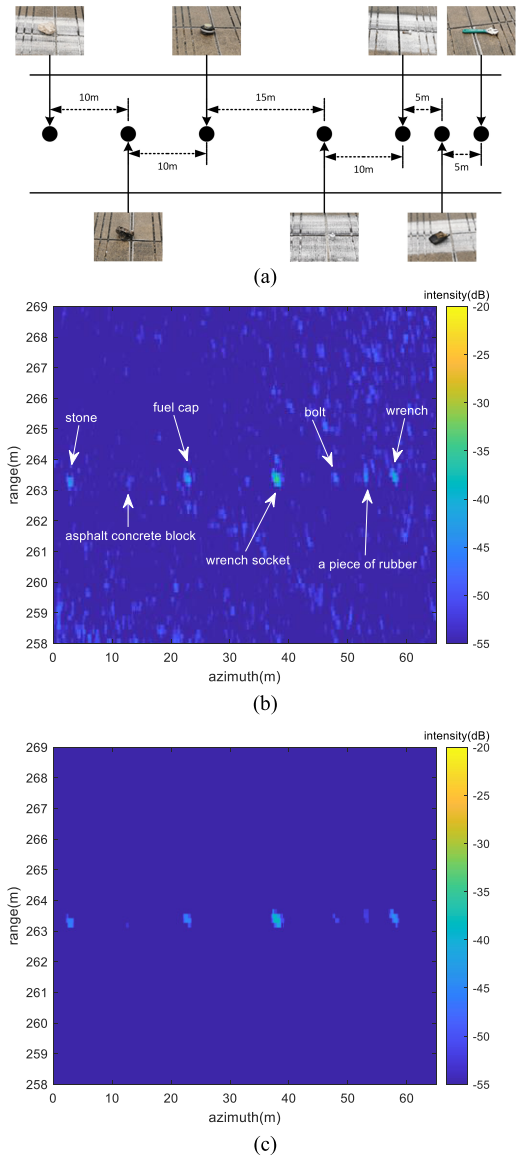


Fig. 20. AS-FODR image of specific FOD items: (a) items' layout of the experimental scene; (b) AS-FODR image; and (c) detection image.

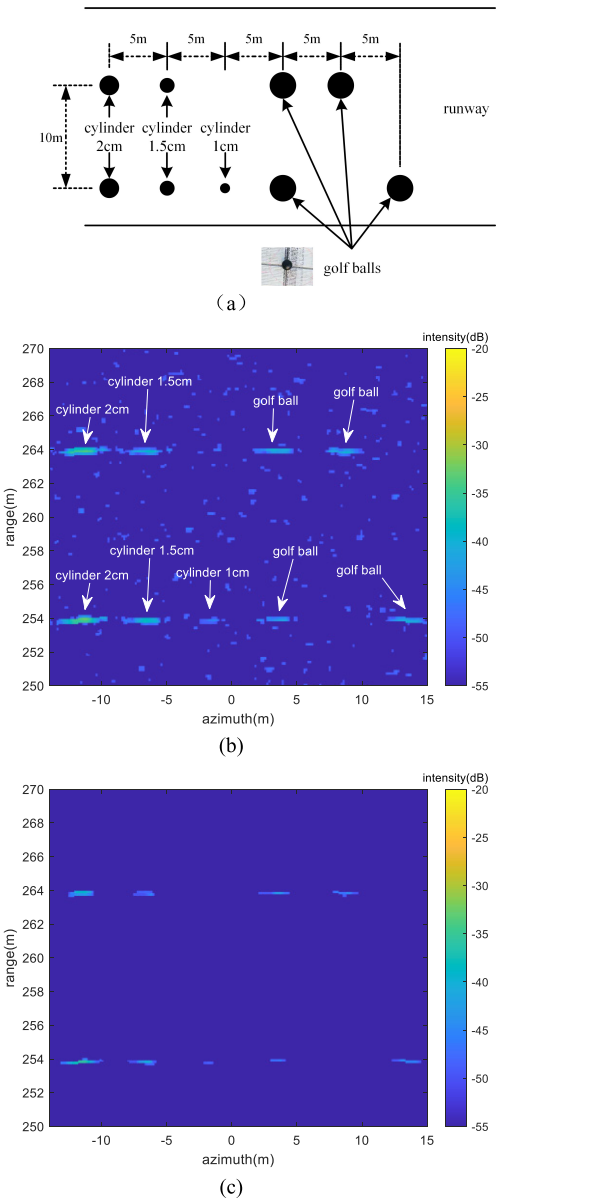


Fig. 21. AS-FODR image of the cylindrical targets and golf balls: (a) targets' layout of the experimental scene; (b) AS-FODR image; and (c) detection image.

experimental test with another AS-FODR prototype at Kaihui general airport in Changsha, Hunan Province on November 8, 2021. The targets' layout in the experimental scene is shown in Fig. 19(a). Starting from 30 m, ten metal cylinders with a diameter of 4 cm are placed at intervals of 5 m. The height of AS-FODR is 0.7 m. Since the 3-dB width of the antenna beam angle (horizontal) is 90°, the target can have an echo response in the entire range of 90°, such as the real aperture image shown in Fig. 19(c), where the azimuth resolution is low. After azimuth compression, the target azimuth resolution is greatly improved, as shown in the synthetic aperture image in Fig. 19(b). It can be seen that azimuth compression can effectively improve the azimuth resolution of the target.

E. AS-FODR Imaging and Detection Results of Other Specific FOD Items

On October 15, 2021, from heavy rain to moderate rain, we carried out a field AS-FODR imaging and detection test on specific FOD items. These items mainly include a stone, an asphalt concrete block, a fuel cap, a wrench socket, a bolt, a piece of rubber, and a wrench. According to the regulations of the FAA and the Civil Aviation Administration of China (CAAC) [8], [49], the length of all these targets in any direction, except the wrench, shall not exceed 10 cm, and the length of the wrench shall not exceed 20 cm. Fig. 20(a) displays those items' layout of the experimental scene, which are placed on the runway and are more than 260 m away from the radar. The imaging result is shown in Fig. 20(b). It can be seen that the electromagnetic scattering intensity of asphalt concrete is equivalent to that of a 1-cm metal cylinder, and the electromagnetic scattering intensity of other specific FOD items is equal to or exceed the intensity of a 1.5-cm metal cylinder. The corresponding detection image is given in Fig. 20(c). Despite the influence of rainfall, specific FOD items can be extracted effectively.

On June 23, 2021, we carried out a field AS-FODR imaging and detection test on golf balls on a sunny day. A golf ball is a standard FOD item specified by the FAA and CAAC.

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The diameter of the golf ball is 4.3 cm. Fig. 21(a) displays the layout of golf balls and metal cylinders of the experimental scene, which are placed on the runway and are more than 250 m away from the radar. The imaging result is shown in Fig. 21(b). Through testing, it is found that the electromagnetic scattering intensity of a golf ball is between that of a 1.5-cm metal cylinder and a 2-cm metal cylinder. The corresponding detection image is given in Fig. 21(c); four golf balls can be extracted effectively.

In summary, the AS-FODR can image and detect FOD items used in two experimental tests over a long range.

VI. Conclusion

In this article, a novel FOD detection radar technology based on AS-SAR is proposed; this technology is superior to the existing FOD radar products in image resolution, direction-finding accuracy, detection performance, and flicker clutter suppression. Additionally, an AS-FODR is designed and implemented with this technology. This system mainly includes the following aspects: the design and processing of the radar engineering structure; the design and analysis of the radar sensor scheme, including frequency selection, system architecture, and working parameters; and the design and implementation of an information processing algorithm flow. According to theoretical and simulation analysis, the AS-FODR can effectively suppress flicker clutter while still achieving a better detection performance under heavy rainfall conditions. Finally, through field experiment tests, the AS-FODR can detect FOD targets on the runway 250 m away from the radar on both sunny and rainy days. Unlike the existing real aperture radar, which scans, images, detects, and reports in real-time, AS-FODR has richer azimuth information, but it can image, detect, and report only after the scanning is completed. At the same time, AS-FODR has more complex information processing, higher requirements for system phase consistency, and higher requirements for computing resources. The AS-FODR algorithms often need to be accelerated by GPU and other hardware. In the future, to improve the detection performance of the AS-FODR, it is necessary to further study superresolution imaging technology and target scattering characteristics.

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Yuming Wang (Member, IEEE) was born in Huining, Gansu, China, in 1985. He received the Ph.D. degree in information and communication engineering from the School of Electronic Science and Engineering, National University of Defense Technology, Changsha, China, in 2013.

Since 2019, he has studied small target detection based on arc-scanning synthetic aperture radar (SAR), which can reduce the threat of airport runway foreign object debris (FOD). His research interests include SAR and its signal processing, image processing, target detection, and machine learning.



Qian Song was born in Wuhan, Hubei, China, in 1973. He received the B.S. and Ph.D. degrees in electronics engineering from the National University of Defense Technology (NUDT), Changsha, China in 1995 and 2001, respectively.

From 2001 to 2003, he was a Post-Doctoral Researcher with the Institute of Microelectronics, Tsinghua University, Beijing, China. From 2004 to 2018, he was an Associate Professor with the School of Electronics Science and Engineering, NUDT, where he has been a Professor since 2012. He is currently the Executive Director of Hunan GHz Info. Tech. Company Ltd., Changsha, a Chinese company he founded in 2019 to develop and commercialize mobile slope stability monitoring radar and airport foreign object debris (FOD) detection radar. He has authored two books, more than 100 articles, and more than 30 inventions. His research interests include highresolution synthetic aperture radar (SAR) imaging, ground-based interferometric SAR, and weak target detection in cluttered environments.



Jian Wang received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2003, 2004, and 2009, respectively.

He is currently a Lecturer with the College of Electronic Science, NUDT. His research interests include ground surveillance radar, target tracking, synthetic aperture radar (SAR) image formation, and SAR interferometry.



Huimin Yu received the B.E. degree in electronic engineering from Hunan Normal University, Changsha, China, in 2003, and the Ph.D. degree in electromagnetic fields and microwave technology from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2009.

He is currently an Associate Professor with the Department of Communication Engineering, Hunan Normal University. His research interests include wireless communication and ultrawideband system design.

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